

Part 1 of 2

An Introduction to Wireless Sensor Networks

Wireless sensor networks (WSNs) have seen explosive growth in recent years because of their independent sensing capabilities. Study of WSNs requires knowledge of many subjects. First part of his article gives basics of the technologies used in WSNs. The next part will cover network protocols

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A wireless sensor network (WSN) is a communication system which senses and gathers information from a certain area and sends it to different places of its choice. In other words, WSN connects physical world to virtual world. In such networks the communication system requires networking protocols that are efficient, reliable, scalable and secure.

WSNs can be very useful for target detection and identification by military, and for environmental applications like agricultural farm monitoring and forest fire monitoring. Sensors integrated into structures, machinery and the environment, coupled with efficient delivery of sensed information, could provide tremendous benefits to society, such as fewer catastrophic failures, conservation of natural resources and improved manufacturing productivity.

These applications require a good quality of service (QoS) from sensor networks, such as, minimum percentage of sensor coverage in the required area, continuous service during required time slot with minimum (or limited) resources (like sensor energy and channel bandwidth) and minimum outside intervention. The sensors may vary from miniature in-built sensors to external sensors like video cameras and position sensors.

WSNs have witnessed tremendous growth recently because of their location-independent sensing capabilities even in inaccessible and dangerous

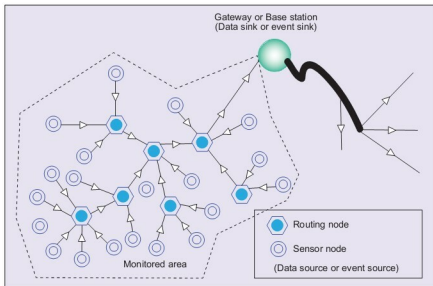


Fig. 1: Monitored area with WSN network

regions, low cost of sensors and long life of power supply because of low power consumption.

The WSN is a variant of ad-hoc networks and differs from ad-hoc networks because of its constraints and unique features. Ad-hoc network means each sensor supports a multi-hop routine algorithm (several nodes may forward data packets to the base station). The WSN lifetime is dependent on the available energy, so energy routing protocol is desired, which can significantly prolong the life-time of WSN. A WSN requires collaboration among multiple types of sensors with different specifications, thus forming a heterogeneous WSN.

Wireless technologies aim at achieving ambient intelligence. Mobile devices embedded in homes, offices and public

places enable ubiquitous sensing, computing and communication. Mobility also implies that limited size and restriction on power consumption, seamless connectivity with other devices and fixed networks, and increased data rates are the bare minimum necessities for intelligent systems.

Features of WSNs

Following are some unique features of WSNs:

1. Sensor nodes are typically immobile
2. WSNs may be deployed in harsh environments, so failures may be quite common
3. The WSNs may be very small, thereby requiring smaller batteries which are bound to have short life-time.